

Should States Encourage Federal Benefit Takeup?

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Abstract

States often control the process for claiming benefits that the federal government largely finances. I ask how this division of responsibility changes the marginal value of public funds (MVPF) for the state making the participation decision. The analysis distinguishes the return to the state and local treasury from the value to state residents, who bear a share of federal costs. I retain each paper's causal estimates and willingness-to-pay convention, then assign fiscal components to their payers when the available sources identify them. In a Pennsylvania SNAP experiment, two consolidated MVPFs below one become state-resident values close to ten. California's move to semiannual SNAP reporting pays for itself for residents under the included components if enough of the assumed administrative saving can be realized. A reconstruction of Virginia's CommonHelp portal, calibrated to the authors' figure, gives a state-resident benchmark of 4.60 compared with their consolidated estimate of 1.27. Fiscal incidence changes the denominator, while the value that marginal participants place on enrollment remains a separate question.

1. Introduction

States make many of the practical decisions that determine whether an eligible person receives a transfer, including the application form, recertification schedule, reminder letters, online portals, and sometimes the default following an eligibility determination. Yet the federal government may pay most of the resulting benefits while a state or county bears the cost of changing the application process. How, then, should a state evaluate a policy that makes a federally financed benefit easier to claim?

The conventional MVPF combines fiscal effects across all levels of government, which makes it the appropriate national benchmark. A state deciding whether to change its own administrative policy faces a narrower account because state and local governments bear the costs and receive the savings recorded in their budgets, while residents finance only a share of federal spending. I work out this account and apply it to policies intended to increase or maintain benefit participation.

The Pennsylvania SNAP experiment studied by Finkelstein and Notowidigdo (2019) shows how large the difference can be. Information sent to likely eligible older adults has a consolidated MVPF of about 0.89, while adding application assistance produces an estimate of about 0.91 after including the resources used to provide it. Counting Pennsylvania's costs in full and allocating

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federal costs by the state's population share gives state-resident values of 10.57 and 9.78, with corresponding state and local treasury values of 19.25 and 16.35.

Each calculation answers a different question: the national calculation counts the full federal outlay regardless of where recipients live. The treasury calculation compares welfare accruing to Pennsylvania residents with the cost recorded in Pennsylvania state and local accounts, leaving federal benefit payments outside its budget boundary. The state-resident calculation adds the portion of federal spending ultimately borne by Pennsylvania residents. Federal financing is therefore inexpensive from the perspective of a small state, although residents still bear some of its cost through the federal fiscal system.

Take-up policies make this separation especially important. A state can mail a letter that induces a federal SNAP payment, while a county can remove a recertification step that saves worker time and extends federally financed benefits. A common application portal may change participation in several programs, each with different benefit rules, administrative match rates, and longer-run fiscal consequences. The relevant federal share must therefore be determined separately for each marginal component of the policy. An average program match rate cannot establish whether a new activity qualified for reimbursement, whether the relevant grant was capped, or whether an administrative saving could be removed from the agency's budget.

I capture these distinctions with three accounting boundaries. Let W_N denote the welfare value in the consolidated calculation and W_s the portion accruing to residents of state s . Let $G_{SL,s}$ be the net cost to the state and its local governments, $G_{F,s}$ the associated federal net cost, and $G_{O,j}$ a cost included in another government or outside account. Costs enter as positive numbers and savings as negative numbers, giving

$$M_N = \frac{W_N}{G_{SL,s} + G_{F,s} + \sum_j G_{O,j}}, \quad M_{T,s} = \frac{W_s}{G_{SL,s}}, \quad M_{R,s} = \frac{W_s}{G_{SL,s} + \lambda_s G_{F,s} + \sum_j \omega_{sj} G_{O,j}}. \quad (1)$$

The first ratio is the conventional MVPF with government accounts consolidated, the second uses the state and local treasury boundary, and the third is the state-resident MVPF. Here, λ_s is the share of a marginal federal fiscal dollar borne by residents of state s , and ω_{sj} assigns the analogous incidence for another outside component. Setting these weights to one returns the consolidated denominator when the component ledgers are the same; recovering the full conventional MVPF also requires the resident and national welfare numerators to coincide.

Consider a one-dollar increase in federally financed SNAP benefits paid to a resident. The treasury denominator omits the dollar because it never enters the state budget, the state-resident denominator counts λ_s dollars, and the consolidated calculation counts the whole dollar. The treatment effect and benefit received remain the same across the three accounts, while the share of fiscal cost entering the denominator changes.

For λ_s , I report the state's population share and its share of federal individual income taxes after credits as two reference values. Their construction can be observed, and comparing them shows whether the result depends on the choice of state-size measure. Neither measures the incidence of a

marginal federal dollar, which could be financed through payroll taxes, corporate taxes, borrowing, or a reduction in some other expenditure. When included welfare is positive and the denominator is zero or negative, I report that the policy pays for itself under the included components. A ratio would give a fiscal saving an unfavorable sign, or become arbitrarily large as the denominator approached zero.

The empirical exercise reallocates published welfare accounts without reestimating the policies' causal effects. I preserve each paper's treatment effect and willingness-to-pay convention where possible, then assign benefit spending, administrative costs, fiscal externalities, and implementation resources to their payers when the available evidence identifies them. Where it leaves the payer unidentified, I report scenarios.

Four labels distinguish the evidentiary status of individual inputs and calculations. An input is *verified* when it appears in the paper or an authoritative program source, and *derived* when it follows arithmetically from those inputs. It is *illustrative* when a missing incidence quantity requires an explicit assumption, and *open* when public evidence leaves that quantity unidentified. The labels describe particular inputs and calculations, with no assessment implied for a paper as a whole.

Hendren and Sprung-Keyser (2020) consolidate government accounts in their conventional MVPF calculation, so moving a dollar between them creates no social value by itself. I retain that calculation as the national benchmark and ask how its components enter the decision of a government representing a narrower constituency. Changing the boundary reallocates the existing denominator by fiscal incidence and restricts the numerator to resident welfare; it creates no new welfare effect. The state-resident value serves decentralized policy choice, while the national value remains the appropriate measure for questions about national efficiency.

Agrawal, Hoyt, and Ly (2024) study a related distinction in a broader model of decentralized policy. Their local MVPF places the willingness to pay of the enacting jurisdiction's residents over the net cost on that jurisdiction's budget, so it corresponds most closely to the treasury MVPF in equation (1). Their social MVPF adds welfare and fiscal effects throughout the federation and is the natural analogue of my national benchmark. The state-resident MVPF makes a separate incidence adjustment because residents bear part of federal financing even when the expenditure never appears in the state budget. I use these definitions to locate the accounting boundaries for take-up policies and leave their broader model of decentralized policy outside the analysis.

Notice that fiscal reallocation leaves several difficult questions in take-up analysis unresolved. It cannot tell us whether a nonparticipant lacked information, whether an application burden used real resources, or how a marginal recipient valued an in-kind transfer relative to its budgetary cost. A large treasury ratio also supplies no general verdict on the policy's national desirability.

I begin with Pennsylvania, where a randomized change in participation gives a finite benchmark, and then consider California, where assumed administrative savings can reverse the denominator's sign. The third main case is Virginia's CommonHelp portal, whose several programs and payers support informative scenarios without delivering a unique state-resident estimate. I close with shorter cases in which match rates, the policy boundary, or implementation costs remain unresolved.

2. Pennsylvania SNAP: a finite benchmark

Pennsylvania provides the most direct illustration of how the fiscal boundary changes the calculation. Finkelstein and Notowidigdo (2019) studied Medicaid recipients age 60 or older who were thought likely to qualify for SNAP but were not receiving it. Pennsylvania randomly assigned them to a control group, an information treatment, or an information-plus-assistance treatment. The information group received a letter explaining that they might qualify, a telephone number for the Pennsylvania Department of Human Services, and ordinarily a reminder postcard. The assistance group received nearly identical information and a referral to Benefits Data Trust (BDT), which screened callers, filled out and submitted forms, helped collect documents, and resolved problems after submission.

Both treatments increased participation, with information raising applications by 7.0 percentage points and enrollment by 4.7 points, and assistance raising them by 16.1 and 11.8 points. With only 5.8 percent of the control group enrolling, assistance roughly tripled enrollment relative to control. Outreach also changed who entered: marginal enrollees were less sick and had higher incomes than existing recipients, and they disproportionately received the \$16 minimum benefit. This selection means that welfare cannot be calculated by valuing every induced benefit dollar at one dollar.

Finkelstein and Notowidigdo infer that marginal applicants had substantially underestimated SNAP's value. Their behavioral calculation values information by the gap between expected benefits and the value revealed by prior nonapplication, while the assistance calculation also includes reduced private application burden. Applying the paper's type-specific belief errors, benefit amounts, and application effects gives welfare of \$199.152 per targeted person for information and \$466.8384 for assistance. Of the latter amount, \$17.85 comes from eliminating a \$75 application cost for each of the 23.8 percent of targeted people who apply.

For the same targeted population, information induces \$205.20 in expected benefits and \$18.69 in processing, or \$223.89 before delivery, and approximately \$1 for the letter and postcard brings total cost to \$224.89 and the consolidated MVPF to 0.886. Assistance induces \$462.78 in expected benefits and \$42.72 in processing, producing a ratio of 0.924 when assistance itself is assigned no resource cost. BDT's estimated cost adds \$7.20 per targeted person, raising total cost to \$512.70 and yielding an MVPF of 0.911. I use this resource-inclusive value for the implemented policy because BDT staff time remains costly even when the state does not write the check.

Under the financing rules in place in 2016, the federal government paid the full cost of SNAP benefits, while Pennsylvania and the federal government divided ordinary eligibility administration approximately equally. Available documents leave the reimbursement of the experimental mailings and BDT services unresolved, so I conservatively assign both to Pennsylvania. Under this assignment, information costs Pennsylvania \$10.345 and the federal government \$214.545, while assistance costs them \$28.560 and \$484.140, respectively. Notice that one generic federal share cannot be applied to the total because benefits, routine processing, and delivery each follow a different financing rule.

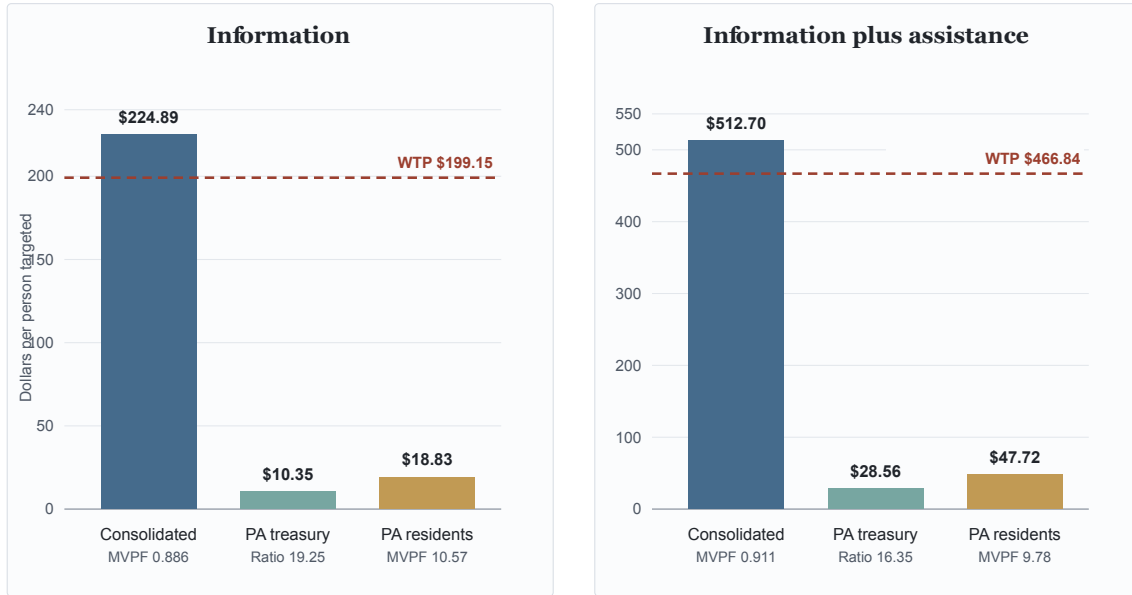
The resident calculations are therefore

$$M_{R,PA}^I(\lambda) = \frac{199.152}{10.345 + 214.545\lambda}, \quad M_{R,PA}^A(\lambda) = \frac{466.8384}{28.560 + 484.140\lambda}. \quad (2)$$

Pennsylvania accounted for 3.956 percent of the US population in 2016, so $\lambda = 0.03956$ gives 10.57 for information and 9.78 for assistance. Using Pennsylvania's share of federal individual income tax after credits gives 10.68 and 9.87, a very similar pair of estimates. The treasury limit, $\lambda = 0$, yields 19.25 and 16.35, while $\lambda = 1$ returns the consolidated values of 0.886 and 0.911. Finkelstein and Notowidigdo do not report these state figures, which are my calculations under the stated financing and incidence assumptions.

Alternative delivery assignments change the level while preserving the basic result, with a 50/50 division of routine processing and experimental outreach or BDT resources giving population-share values of 10.85 and 10.55. Assigning all processing and delivery costs to Pennsylvania instead gives 7.16 and 6.84. Every denominator remains positive, and Appendix B separates the cases because eligibility for reimbursement does not establish that reimbursement was requested or received.

The fiscal denominator changes with the accounting boundary



Pennsylvania-resident bars use the 2016 population-share federal-incidence benchmark.

Figure 1. Each panel holds recipient willingness to pay fixed and changes only the fiscal boundary. The Pennsylvania-resident bars use the state's 2016 population share as the federal-incidence weight. Dollar amounts are per person targeted. Source: author's calculations from Finkelstein and Notowidigdo (2019) and its online appendix.

Why does an MVPF below one become a value near ten? The \$199.152 information benefit accrues to Pennsylvania residents, whose denominator includes the full state cost and only \$8.49 of the \$214.545 federal cost because they finance a small share of federal spending, giving a denominator of \$18.83. Assistance generates substantially more enrollment, yet its additional processing and

BDT resources produce a \$47.72 denominator and an MVPF of 9.78, slightly below the information value.

Both denominators are positive, so unmeasured health or tax savings would have to exceed \$18.83 or \$47.72 before the respective policy paid for itself on the included components, while fixed management, data, call-center, or IT costs would increase them. The paper supplies no policy-specific amount for these items, and their payer remains open.

The numerator also carries substantial uncertainty because random assignment identifies the enrollment effects, while prior beliefs and willingness to pay are inferred. The \$75 burden reduction assumes five hours valued at \$15 per hour, although recipients still had to complete the required state interview. The \$267 processing-cost input reproduces the published welfare calculation but does not transparently reconcile with a nearby \$134 administrative statistic. Burden enters willingness to pay, while processing and implementation costs enter the fiscal denominator, and neither assumption changes the randomized participation effect.

3. California CalFresh: when simplification saves money

California’s reporting reform makes the resident denominator negative because the calculation includes substantial administrative savings. In October 2013, the state replaced the quarterly eligibility reports filed by most CalFresh households with semiannual reports, eliminating the report due around the third month after enrollment. Unrath (2024) compares cohorts entering immediately before and after the change and finds that six-month retention rose from 78.3 to 87.0 percent, or 8.7 percentage points. This adjacent-cohort comparison displays neither an untreated group nor sampling uncertainty, and the paper’s text and figure note use different pre- and post-reform windows. Its participation estimate is therefore less secure than Pennsylvania’s randomized estimates.

The March 2024 paper reports a consolidated MVPF of 4.25 for a policy unit containing one high-food-security and one low-food-security case. Over months four through six, the reform produces \$138.16 in additional benefits and removes \$42.02 in private reporting burden for marginal and inframarginal recipients, yielding \$180.18 in willingness to pay. Government incurs the same \$138.16 in benefits and a \$9.26 fiscal externality imported from other studies, while the calculation assumes \$105.05 in avoided processing. Net government cost is \$42.37, so $180.18/42.37 = 4.25$.

Historical financing places these components in different accounts. CalFresh benefits were federal, while ordinary administration was 50 percent federal, 35 percent state, and 15 percent county. The \$9.26 fiscal externality combines an assumed adult labor-supply cost with a future child-earnings saving, both imported from other studies, and neither Unrath nor the underlying sources allocates it across governments. Let q_K denote the share assigned to California state and local governments, and let λ denote the share of federal cost borne by California residents. The resident denominator is

$$D_{CA}(\lambda, q_K) = -52.525 + 9.26q_K + \lambda(94.895 - 9.26q_K). \quad (3)$$

The -52.525 term is the state and county share of the assumed administrative saving, and the next term assigns part of the fiscal externality to California. The expression in parentheses contains

federal benefits, federal administrative savings, and the remaining externality. I retain q_K because the evidence provides no basis for selecting an allocation.

State and local governments save \$52.53 per normalized unit when none of the fiscal externality falls on California and \$43.27 when all of it does. Adding federal cost attributable to California residents leaves the denominator negative under both incidence measures. The range is -\$41.02 to -\$32.88 using population and -\$39.44 to -\$31.46 using federal income tax after credits. Under the paper’s component assumptions and this manuscript’s financing and incidence scenarios, the policy therefore pays for itself for California residents. A negative MVPF would give a misleading sign to positive welfare accompanied by fiscal savings.

This conclusion depends on realizing the assumed administrative saving. Unrath assigns \$50 in public cost to every report and applies the saving to marginal and inframarginal cases, although the paper estimates neither California’s marginal processing cost nor the share of worker time converted into lower county or state spending. Under the q_K and λ combination yielding the largest resident cost, the denominator reaches zero when 47.4 percent of the assumed \$105.05 saving is marginal and cashable. A realized share above 47.4 percent leaves the denominator negative, and the available evidence cannot determine whether California cleared that threshold.

The calculation also omits transition costs for systems work, notices, and training. State fiscal documents establish that these activities occurred, but their totals combine several programs and provisions and cannot be mapped into Unrath’s two-case normalization. Other calibrated inputs include the \$20 private reporting cost, which does not follow from the printed time-and-wage calculation, and adult and child fiscal effects drawn from national SNAP studies instead of the California reform.

If California bore all transition costs, \$31.46 to \$41.02 per normalized unit would exhaust the resident fiscal saving across these cases. An equal federal and California split raises the corresponding range to \$55.29 to \$73.17. Because the aggregate fiscal notes omit the exposure and amortization terms needed to express statewide totals on this scale, these bounds give the transition costs that would reverse the sign conditional on the rest of the calculation.

4. Virginia CommonHelp: several programs and several payers

Virginia opened CommonHelp statewide in October 2012, allowing households to apply online for SNAP, Medicaid and FAMIS, TANF, child-care assistance, and energy assistance, with online recertification added later. For households with children, Cholli and Wu (2026) use distance to social-service offices and discontinuities at office-assignment boundaries to estimate the response to this lower administrative burden. Multiple-program participation rose by 4.183 percent, while single-program participation fell by 1.732 percent. Because the eligible population for each program combination is unavailable, these estimates describe caseload composition and cannot identify take-up among verified eligible households.

Cholli and Wu report a consolidated MVPF of 1.27 for the complete policy. Their calculation values time and travel saved during application and recertification, additional program receipt, the \$10.78 million development cost, changes in benefits and variable administration, and several fiscal externalities. A large externality assumes one dollar of long-run government medical savings for

every dollar spent on childhood Medicaid. The paper reports Medicaid and SNAP separately as well, although the complete-policy estimate requires a joint calculation because one portal changed households' combinations of benefits.

Several financing rules determine how much of the resulting cost belongs to Virginia, since SNAP benefits were federally financed while administration was shared. Medicaid and FAMIS had different match rates, with the composition of marginal child enrollment unavailable. TANF combined a fixed block grant with state maintenance-of-effort spending, so an accounting share may differ from the opportunity cost of another dollar. Virginia and its local departments also shared administrative responsibilities, and several sources financed the broader eligibility system. The government receiving future medical savings depends on the child's residence, subsequent coverage, and public payer.

The public paper provides a complete numerical decomposition only for development spending; other inputs appear as bars in Figure A.32, without a numerical workbook or replication code for the welfare exercise. I use approximate graph-calibrated anchors of \$63.10 million in total welfare value and \$85.30 million in direct government cost. The residual fiscal externality required to reproduce the authors' exact 1.27 is -\$35.615 million, leaving consolidated net cost of \$49.685 million. Cholli and Wu report the consolidated result, while the component anchors and state calculations are my derived scenarios calibrated to Figure A.32.

My benchmark assigns \$12.758 million of net cost to Virginia state and local accounts and \$36.927 million to a combined federal and outside-government account. I apply historical program rules where possible, assign none of the modeled \$37.1 million child-Medicaid saving to Virginia, split portal development equally between Virginia and federal accounts, and assign Virginia 20 percent of the small remaining tax-and-transfer externality. Using Virginia's population share for incidence gives a resident denominator of \$13.723 million. The complete-policy MVPF is 1.27 under consolidated accounting, 4.95 at the treasury limit, and 4.60 for Virginia residents; Virginia's federal-income-tax share gives 4.56. I use 4.60 as the benchmark, while recognizing that one λ combines several distinct federal and outside-government payers.

A direct-budget sensitivity removes all modeled behavioral fiscal externalities, uses a different component ledger, and gives 4.39. The benchmark is conservative about the long-run medical saving because it assigns none of that saving to Virginia. Other allocations can move the estimate in either direction through the TANF opportunity cost, portal financing, or the division of marginal Medicaid and FAMIS enrollment. A later report on Virginia's broader eligibility system assigns 18.7 percent of development spending to the state general fund. Applying that share to CommonHelp and treating the TANF residual as fully controlled by Virginia raises the resident ratio to about 7.26 and lowers the medical-saving crossing to about 24 percent. Because that report covers more than the 2012 portal and a different period, I use it only as a sensitivity. It suggests that a 50/50 development split may assign Virginia too much cost, while the portal's original marginal financing remains unknown.

The largest variation comes from the jurisdiction of the long-run medical saving. Let s denote Virginia's share of the modeled \$37.1 million saving, with the remainder assigned to the combined federal and outside-government account. Under population-share incidence, the resident denominator is

$$D_{VA}(s) = 13.723 - 37.1(1 - 0.02613)s. \quad (4)$$

The factor $1 - 0.02613$ reflects that Virginia receives a dollar assigned to its accounts in full, while a dollar left outside enters with the state’s population share. Assigning one-quarter of the saving to Virginia lowers the denominator to \$4.690 million and raises the ratio to 13.45. It reaches zero at approximately $s = 0.38$, after which the policy pays for itself under this scenario.

The 38 percent crossing describes fiscal jurisdiction: it is the share of the modeled saving Virginia would need for its resident denominator to reach zero, without predicting actual receipt. The evidence follows children in a different setting and extrapolates medical utilization through age 65, while future residence and the public payer remain unknown, so I carry forward the 4.60 benchmark and the 38 percent crossing.

5. Additional participation cases

Each of the four remaining studies leaves a match rate, policy boundary, or implementation cost unresolved, although three evaluate rules that change enrollment or retention directly. The fourth estimates caseworker effects without evaluating a policy that would generate the observed improvement; Appendix E provides the component calculations.

Shepard and Wagner (2025) study automatic assignment to a plan in Massachusetts Commonwealth Care for applicants below poverty who had already been found eligible and failed to choose one. Suspending assignment reduced new enrollment by 32.6 percent relative to its automatic-enrollment baseline, so adding it increased enrollment by 48.4 percent relative to active enrollment alone. The consolidated welfare ratio is approximately one. My resident calculation assigns 36.5 percent of uncompensated care to government and 63.5 percent to providers, gives the government offset the coverage match, uses a 3.228 percent federal-incidence weight, and sets unmeasured implementation cost to zero. Because provider ownership and owners’ residence are unavailable, their Massachusetts share varies from zero to one. The resident ratio is 1.01 to 1.94 at a 50 percent match and 1.29 to 2.48 at 61.59 percent. These ranges also reflect financing-regime changes across enrolled months and an unknown marginal public offset for uncompensated care.

Gray et al. (2023) examine Virginia’s historical three-month time limit for able-bodied adults without dependents, whose elimination increases participation and changes substantive eligibility and work conditions along with recertification burden. The consolidated MVPF is 0.90 to 1.40. Under historical financing rules, elimination saves the Commonwealth and local governments \$1.21 per relevant ABAWD-month on measured components. Using the authors’ central earnings calibration over 24 months, zero net Virginia income tax for the representative filer, published WTP endpoints, and both population and federal-income-tax shares to allocate federal costs produces resident MVPFs from 64.56 to 118.08. Their denominators are only \$0.99 to \$1.15, so a modest omitted transition cost or another earnings calibration matters more than the difference between 65 and 118.

Cook and East (2026) study SNAP parents in an anonymous state whose youngest child turned six shortly before recertification, making the household head subject to referral for mandatory

Employment and Training. Eliminating mandatory E&T restores an estimated \$111.50 in monthly SNAP benefits per induced referral and subtracts \$11.32 in planned average E&T cost, yielding a consolidated MVPF of 1.11. E&T could be financed from a finite grant with a 100 percent federal share, 50/50 reimbursement, or an unidentified nonfederal payer, so the paper does not identify a state-resident point. An illustration with federal incidence of 0.02 treats supervised job search as formula-funded, assumes released formula dollars reduce federal outlays dollar for dollar, assigns all nonfederal reimbursement savings to state residents, and sets omitted fiscal and transition effects to zero. It gives 103.95. Under 50/50 funding, the denominator can become nonpositive, making that sign change more informative than a ratio that diverges near zero.

Cook and East (2025) use quasi-random assignment to SNAP caseworkers in an anonymous mountain-plains state. Assignment to a caseworker one standard deviation more effective at producing receipt raises initial receipt by 2.68 percentage points and reduces incomplete applications by 1.27 points. Their hypothetical welfare calculation for this improvement is 3.3, while the unrounded displayed inputs give 3.13. The study shows that caseworkers affect outcomes after submission, yet it leaves the policy for improving effectiveness unspecified. Training, staffing, monitoring, caseload changes, and software could each produce different effects at different costs, and the paper measures none of these technologies. Because the paper identifies neither that implementation technology nor its cost or state, it cannot support an implementable state-resident MVPF.

6. Lessons for state policy

A state evaluating a participation barrier needs evidence on three separate margins, beginning with the participation response. Pennsylvania estimates it through random assignment, California uses a less secure adjacent-cohort comparison, and CommonHelp combines geographic and office-boundary designs. Valuing the marginal participants then requires a view about misinformation, application ordeals, transfers, or future benefits, while fiscal evaluation adds each payer’s identity and federal-cost incidence. A strong design can identify participation while leaving the payer unknown, and a high federal match contains no information about recipient value.

Program labels give little guidance about marginal cost because Pennsylvania combines federally financed benefits, 50/50 processing, and delivery resources conservatively assigned to the state. California’s denominator changes sign through assumed state and county administrative savings, while CommonHelp combines open-ended benefits, matching administration, a fixed block grant, portal development, and medical savings that another jurisdiction may receive. Eligible spending may go unreimbursed, average federal shares can misstate costs when a grant cap binds, and an hour of worker time saves cash only if staffing or another paid input falls. Statutory shares begin the calculation, while each component still needs a marginal financing analysis.

The component ledger is therefore as important as the ratio and should show resident willingness to pay, state and local net cost, federal net cost, outside fiscal components, and the incidence weights joining them. Scenario ranges track an unknown payer or normative parameter, while sampling uncertainty measures precision in the underlying estimate. This distinction matters most near zero, where a one-dollar denominator change can turn a large finite MVPF into a fiscal saving. Reporting the denominator and its zero crossing makes clear why an MVPF of 100 on one dollar of cost is

less stable than an MVPF of ten on fifty dollars. The ledger should also bound implementation, transition, and IT costs when direct estimates are unavailable.

Taken together, these calculations show why state take-up policies can look much more attractive to state treasuries and residents than under consolidated accounting. Pennsylvania produces finite resident values near ten because the federal government pays most induced benefits. California pays for itself on the included components once enough administrative saving is realized, while CommonHelp’s 4.60 benchmark depends on who receives future medical savings. Determining which result applies requires assigning every component to a level of government, checking whether reimbursement was claimed, and measuring marginal administrative and implementation costs. Where the evidence runs out, the denominator and its zero crossing show how large the missing input must be to change the conclusion.

Appendix A. Accounting definitions and conventions

A.1 The three accounting boundaries

For a policy implemented in state s , let W_N be the welfare numerator in the national calculation and let W_s be the portion accruing to residents of that state. I use $G_{SL,s}$, $G_{F,s}$, and $G_{O,j}$ for the net fiscal costs borne by state and local government, the federal government, and an outside account j . Costs are positive and savings are negative. With this notation, the three objects in Equation (1) are

$$M_N = \frac{W_N}{G_{SL,s} + G_{F,s} + \sum_j G_{O,j}}, \quad M_{T,s} = \frac{W_s}{G_{SL,s}}, \quad M_{R,s} = \frac{W_s}{G_{SL,s} + \lambda_s G_{F,s} + \sum_j \omega_{sj} G_{O,j}}.$$

The treasury-limit ratio asks how much welfare accrues to state residents per dollar of state and local budget cost. Fiscal accounts elsewhere do not enter this denominator, so the ratio should be read as a welfare-over-budget statistic. The state-resident calculation then adds the shares of outside costs that ultimately fall on residents. In the empirical work, I vary λ_s using population and federal individual-income-tax shares. I specify ω_{sj} separately when an outside component has an incidence that differs from federal spending.

The three ratios have a useful relationship in the special case with one state and local account and one federal account. Suppose they have the same numerator and these two accounts exhaust the denominator. Since

$$G_{SL,s} + \lambda G_{F,s} = (1 - \lambda)G_{SL,s} + \lambda(G_{SL,s} + G_{F,s}),$$

it follows that

$$\frac{1}{M_{R,s}(\lambda)} = \frac{1 - \lambda}{M_{T,s}} + \frac{\lambda}{M_N}. \tag{A.1}$$

Thus, the inverse resident MVPF is a weighted average of the inverse treasury and national ratios. Notice that Equation (A.1) requires a common numerator and a fixed, exhaustive two-account ledger. It ceases to describe the relationship once the resident and national numerators differ, an outside account has been omitted, or a sensitivity changes the components themselves. The CommonHelp direct-budget case removes behavioral fiscal externalities, for example. It consequently defines a different ledger and falls outside this interpolation.

A.2 Incidence, signs, and evidence labels

Population shares and income-tax shares provide two transparent rules for allocating a federal cost across states. Neither traces the source of financing for a marginal federal dollar. That dollar could come from current income, payroll, corporate, or excise taxes, from borrowing and future taxes, or from reduced federal spending elsewhere. I use these benchmarks because they are dated and reproducible, and because showing both reveals how much a result depends on the particular measure of state size.

If $W_s > 0$ and the resident or treasury denominator is nonpositive, I say that the policy **pays for itself under the included components and scenario**. A negative or infinite MVPF has little useful interpretation here. When a positive denominator approaches zero, I give both its level and the parameter value at which it crosses zero. This makes clear when a very large ratio is being generated by a denominator close to zero.

I use four evidence labels throughout the appendices:

- **Verified:** printed in the relevant paper or an authoritative program, budget, statutory, or statistical source.
- **Derived:** arithmetic using verified inputs.
- **Illustrative:** a stated value used because an incidence, valuation, or implementation input is unavailable.
- **Open:** not identified by the reviewed public evidence and not silently assigned.

The labels apply to individual claims. Thus, a causal estimate may be verified even though its fiscal allocation is illustrative. Similarly, a statutory financing share may be verified while the amount actually reimbursed in this implementation remains open.

Appendix B. Pennsylvania SNAP component ledgers

B.1 Reconstruction of the two treatments

All amounts in this appendix are dollars per person targeted. In the information arm, the application effects for the high- and low-benefit groups are 0.04 and 0.03. The prose in Finkelstein and Notowidigdo (2019) reverses the subscripts, while its displayed equation and reported composition of marginal applicants use the ordering below. I follow the displayed calculation:

$$W_I = .98(4,806)(.04) + .83(432)(.03) = 199.152.$$

For assistance, the corresponding high- and low-benefit effects are 0.09 and 0.07. These effects imply a behavioral value of \$448.9884 before accounting for application burdens. The source calculation adds \$75 for each of the 23.8 percent of treated people who apply:

$$W_A = .98(4,806)(.09) + .83(432)(.07) + 75(.238) = 466.8384.$$

The following table gives the complete ledger for each treatment. The national column reconstructs the conventional calculation. The Pennsylvania column shows the allocation I use for the state-resident calculation.

Treatment and component	Amount	National ledger	Pennsylvania allocation used here	Status
Information: recipient WTP	199.152	numerator	resident numerator	Derived from printed inputs
Information: induced benefits	205.200	cost	205.200 federal	Derived; benefit financing verified
Information: processing	18.690	cost	9.345 state, 9.345 federal	Derived; historical 50/50 benchmark
Information: mail and reminder	1.000	resource cost	1.000 state	Approximate amount; payer open, allocation illustrative
Information total	224.890	denominator	10.345 state, 214.545 federal	Derived
Assistance: belief-based WTP	448.988	numerator	resident numerator	Derived from printed inputs
Assistance: burden relief	17.850	numerator	resident numerator	Illustrative calibration
Assistance: induced benefits	462.780	cost	462.780 federal	Derived; benefit financing verified
Assistance: processing	42.720	cost	21.360 state, 21.360 federal	Derived; historical 50/50 benchmark

Treatment and component	Amount	National ledger	Pennsylvania allocation used here	Status
Assistance: BDT resources	7.200	resource cost	7.200 state	Derived amount; payer open, allocation illustrative
Assistance total	512.700	denominator	28.560 state, 484.140 federal	Derived

To connect the table to the published estimates, remove the mailing resource from the information denominator. This gives $199.152/223.890 = 0.8895$, which is the published 0.89. Removing BDT resources from the assistance denominator gives $466.8384/505.500 = 0.9235$. The original appendix reports 0.93 and the secondary reconstruction reports 0.92. Once the implementation resources are included, the two ratios are 0.886 and 0.911.

The current Policy Impacts information page displays dollar fields that fail to reproduce the formula on the page, although the resulting ratio still rounds to 0.89. Its assistance page reports a normalized value near 0.92 without showing whether assistance is treated as costless or as a resource cost. I therefore reconstruct both treatments from the article and its online appendix.

One input remains unresolved. The public calculation assigns \$267 of processing cost to each application, and this number reproduces the welfare equation. Nearby, the article describes a cost of approximately \$134 per application, obtained by multiplying \$178 per recipient by a 75 percent acceptance rate. The article and online appendix do not explain how the \$267 and \$134 figures relate. I retain \$267 because it reconstructs the authors' calculation, while treating the provenance of this input as open.

B.2 Federal incidence and financing alternatives

Applying the recommended allocations from Table B.1 gives

$$M_{R,PA}^I(\lambda) = \frac{199.152}{10.345 + 214.545\lambda}, \quad M_{R,PA}^A(\lambda) = \frac{466.8384}{28.560 + 484.140\lambda}.$$

The table traces both ratios as the federal-incidence weight increases from the treasury limit to the consolidated calculation.

Federal-incidence weight λ	Information denominator	Information ratio	Assistance denominator	Assistance ratio
0, treasury limit	10.345	19.251	28.560	16.346
0.020	14.636	13.607	38.243	12.207
0.038705, income-tax share	18.649	10.679	47.299	9.870

Federal-incidence weight λ	Information denominator	Information ratio	Assistance denominator	Assistance ratio
0.039564, population share	18.833	10.574	47.715	9.784
0.050	21.072	9.451	52.767	8.847
0.100	31.800	6.263	76.974	6.065
0.250	63.981	3.113	149.595	3.121
0.500	117.618	1.693	270.630	1.725
1, consolidated	224.890	0.886	512.700	0.911

The population benchmark divides Pennsylvania's July 2016 population, 12,784,227, by the U.S. population, 323,127,513. The income-tax benchmark divides Pennsylvania's 2016 federal individual income tax after credits, \$55.697 billion, by the national total of \$1,439.016 billion. Both remain illustrative incidence proxies, since each allocates the federal cost according to an observed state share.

There is a separate question about the experimental delivery resources. The available evidence does not show which government ultimately paid for the mailing, reminder, and BDT costs. The next table holds welfare and consolidated resource cost fixed and changes only the allocation of those costs.

Allocation	State cost	Federal cost	Population-share ratio	Treasury ratio
Information: processing 50/50, outreach all state	10.345	214.545	10.57	19.25
Information: processing and outreach 50/50	9.845	215.045	10.85	20.23
Information: processing and outreach all state	19.690	205.200	7.16	10.11
Assistance: processing 50/50, BDT all state	28.560	484.140	9.78	16.35

Allocation	State cost	Federal cost	Population-share ratio	Treasury ratio
Assistance: processing and BDT 50/50	24.960	487.740	10.55	18.70
Assistance: processing and BDT all state	49.920	462.780	6.84	9.35

USDA rules permitted federal administrative reimbursement for approved factual outreach and application assistance. This establishes that the expenses could qualify for a match. The public evidence does not show whether Pennsylvania submitted these experimental expenses for reimbursement or received a federal payment. I use the all-state allocation as a conservative upper bound on the state denominator; the actual invoicing remains open.

B.3 WTP and omitted-cost sensitivities

The \$75 value assigned to application assistance is an important valuation choice. Let r denote the burden reduction per applicant. The assistance numerator can then be written as

$$W_A(r) = 448.9884 + .238r,$$

Holding the recommended population-share denominator fixed gives the following sensitivity.

WTP treatment	Information WTP	Information ratio	Assistance WTP	Assistance ratio	Status
Authors' behavioral baseline; full \$75 assistance relief	199.152	10.57	466.8384	9.78	Source calibration plus derived reallocation
No application- burden relief	199.152	10.57	448.988	9.41	Illustrative
Relief only for induced applica- tions, .16 $\times$ 75	199.152	10.57	460.988	9.66	Illustrative

WTP treatment	Information WTP	Information ratio	Assistance WTP	Assistance ratio	Status
Mechanical value equal to induced benefits; full assistance relief	205.200	10.90	480.630	10.07	Illustrative

The calculation may also omit fixed, transition, management, data, call-center, or IT costs. Let k be the amount of any such omitted cost per targeted person paid by Pennsylvania. The resident formulas become

$$\frac{199.152}{10.345 + k + 214.545\lambda} \quad \text{and} \quad \frac{466.8384}{28.560 + k + 484.140\lambda}.$$

At the population benchmark, fiscal savings of \$18.833 per information target or \$47.715 per assistance target would bring the corresponding denominator to zero. The experiment does not estimate a health, tax, or other saving of this magnitude. The state-resident ratios exceed one for $\lambda < 0.8800$ and $\lambda < 0.9053$, respectively. Both denominators remain positive when the calculation is restricted to the measured components.

Finally, these allocations describe the 2016 experiment. Later changes in SNAP administrative or benefit cost sharing do not update the experiment by themselves. A prospective calculation would require a new take-up response, delivery technology, state payment-error status, and reimbursement treatment.

Appendix C. California CalFresh component ledger and thresholds

C.1 Reconstructing the March 2024 calculation

Unrath's Figure 13 is the only internally consistent reconstruction of the reported 4.25:

Component	Amount	Role	Fiscal allocation in the historical benchmark
Additional benefits, high-food-security type	71.94	WTP and cost	federal
Additional benefits, low-food-security type	66.22	WTP and cost	federal

Component	Amount	Role	Fiscal allocation in the historical benchmark
Inframarginal private burden avoided	40.00	resident WTP	California recipients
Marginal private burden avoided	2.02	resident WTP	California recipients
Adult fiscal costs	14.41	government cost	jurisdiction open
Child future-earnings fiscal savings	-5.15	government saving	jurisdiction open
Inframarginal public administration avoided	-100.00	government saving	50 federal, 35 state, 15 county
Marginal public administration avoided	-5.05	government saving	2.525 federal, 1.7675 state, .7575 county
Total	WTP 180.18; net cost 42.37	MVPF 4.25	varies with q_K ; see C.2

Set benefits at $B = 138.16$, private burden relief at $H = 42.02$, the net transported fiscal externality at $K = 9.26$, and administrative savings at $A = 105.05$. Then

$$W = B + H = 180.18, \quad G = B + K - A = 42.37, \quad M_N = 4.2525.$$

The compact equation on page 23 gives WTP of \$182.68, government cost of \$42.2364, and a ratio of 4.33. Its administrative multiplier is 2.14, while the components in Figure 13 sum to $2 + .054 + .047 = 2.101$. The live Policy Impacts page reports an older calculation with WTP \$220.76, net cost \$105.19, and MVPF 2.09. I use Figure 13 in the March 2024 manuscript as the current authoritative ledger.

The private reporting cost contains a second discrepancy. The paper assigns \$20 to two hours valued at twice California's 2013 minimum wage. Since that wage was \$8, the stated formula gives \$32. I retain the operative welfare input of \$20.

C.2 Historical state and federal allocation

Let q_K be the share of the net \$9.26 fiscal externality assigned to California state and local government. During the period studied, benefits were entirely federal and ordinary administration was 50 percent federal, 35 percent state, and 15 percent county. Hence

$$G_{SL} = 9.26q_K - 52.525, \quad G_F = 94.895 - 9.26q_K.$$

q_K	State/local cost	Federal cost	Resident cost, population $\lambda = .121256$	Resident cost, tax $\lambda = .137885$	Result
0	-52.53	94.90	-41.02	-39.44	pays for itself
0.5	-47.90	90.27	-36.95	-35.45	pays for itself
1	-43.27	85.64	-32.88	-31.46	pays for itself

Notice that every state and resident denominator is negative. The population weight divides California’s 2013 population of 38,332,521 by the U.S. population of 316,128,839. The tax weight divides California’s \$169.422 billion in 2013 federal individual income tax after credits by the national \$1,228.719 billion. Even the least favorable assignment, placing all adult cost in California and all child saving in the federal account, leaves resident costs of -\$28.36 and -\$27.02 under the two weights.

C.3 Realizable administration and transition costs

The central calculation treats all \$105.05 of administrative savings as realizable. Let ρ_A be the fraction that is marginal, allowable, and cashable. The resident denominator becomes

$$D_{CA} = q_K K + \lambda[B + (1 - q_K)K] - .5(1 + \lambda)\rho_A A.$$

The minimum ρ_A that makes the denominator nonpositive is:

q_K	Treasury $\lambda = 0$	Population weight	Income-tax weight
0	0.000	0.304	0.340
0.5	0.088	0.373	0.407
1	0.176	0.442	0.474

The least favorable threshold in the central grid is 47.4 percent, though whether California could cash out that share remains unobserved. The paper imports a \$50 processing cost from earlier work and contains no evidence on expenditure reductions or staff reassignment after the report was eliminated.

Let τ be transition, IT, training, notice, and rulemaking cost per normalized policy unit. With a 50/50 split between California and the federal government, the amount that brings the resident denominator to zero is:

q_K	Treasury threshold	Population threshold	Income-tax threshold
0	105.05	73.17	69.32

q_K	Treasury threshold	Population threshold	Income-tax threshold
0.5	95.79	65.91	62.31
1	86.53	58.65	55.29

If California bears the entire transition cost, the corresponding thresholds are \$52.53/\$47.90/\$43.27 for the treasury, \$41.02/\$36.95/\$32.88 with population weights, and \$39.44/\$35.45/\$31.46 with tax weights. Official fiscal documents report material statewide implementation amounts. Because the totals combine CalFresh with other provisions, converting them to the paper’s normalized two-case unit requires an exposure count, a program allocation, and an amortization horizon.

These administrative and transition thresholds concern the denominator. Recipient valuation enters through the numerator. Valuing benefits at one-half of face value reduces WTP to \$111.10 and the national MVPF to 2.62. Valuing only private reporting relief gives WTP of \$42.02 and a national ratio of 0.99. Both sensitivities retain a negative resident denominator and positive WTP in the historical central grid.

Appendix D. Virginia CommonHelp reconstruction

D.1 Graph calibration

Cholli and Wu report an exact complete-policy MVPF of 1.27. Figure A.32 gives its dollar components as unlabeled bars. Reading the figure at 300 dpi gives approximately \$5.5 million of direct WTP, \$58.0 million of indirect WTP, \$85.7 million of direct government cost, and a -\$34.9 million fiscal externality. Given the coarseness of the plot, I use one-decimal anchors of \$63.10 million for total WTP and \$85.30 million for direct cost, then choose the fiscal externality that exactly reproduces 1.27:

$$FE^{all} = \frac{63.10}{1.27} - 85.30 = -35.614961.$$

Consolidated net cost is \$49.685039 million. The dollar components are derived from and calibrated to Figure A.32, while the 1.27 comes directly from the authors.

D.2 Program and payer ledger

The program-payer ledger below is constructed from the graph because Cholli and Wu report only aggregate component bars.

Component, millions of dollars	Amount	Benchmark Virginia/local share	Benchmark federal/outside share	Evidence and principal caveat
Direct WTP	5.30	numerator	n/a	graph anchor, derived

Component, millions of dollars	Amount	Benchmark Virginia/local share	Benchmark federal/outside share	Evidence and principal caveat
Indirect WTP	57.80	numerator	n/a	graph anchor, derived
CommonHelp development	10.779739	50%	50%	spending reported as of launch and treated by the paper as development cost; financing illustrative amount
Medicaid/FAMIS benefits	65.60	47.5%	52.5%	graph-derived; program mix open
Medicaid/FAMIS administration	-3.60	25%	75%	graph-derived; qualifying match open
SNAP benefits	58.50	0%	100%	amount graph-derived; federal financing verified
SNAP administration	-42.00	50%	50%	graph-derived and extrapolated from SNAP spending
TANF direct-cost residual	-3.979739	55%	45%	balancing residual; marginal block-grant incidence open
Medicaid/FAMIS externality	-37.10	0% in benchmark	100% balance	transported long-run saving; jurisdiction open
SNAP externality	1.00	included in 20% rule	balance	graph-derived

TANF/rounding externality	.485039	included in 20% rule	balance	calibrated residual
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The direct-cost rows sum to \$85.30 million, and the externalities sum to -\$35.614961 million. The 47.5 percent medical-benefit share assumes that 15 percent of marginal medical enrollment is FAMIS. Regular Medicaid carried a 50 percent state share, while a time-weighted FAMIS calculation gives 33.23 percent over the evaluation window. The marginal FAMIS fraction is unreported. The 25 percent medical-administration share applies the enhanced federal rate for qualifying eligibility-system operation, whose eligibility has not been established for every affected dollar.

SNAP benefits were federally financed, with administration generally shared. TANF used a fixed federal block grant and state maintenance-of-effort spending. I use a 55 percent state share to approximate Virginia's reported basic-assistance accounting mix, though the opportunity cost of a marginal TANF dollar remains unresolved. CommonHelp also operated through local departments of social services. I combine Commonwealth and local accounts because I could not locate a local allocation for the causal period.

The benchmark direct Virginia/local cost is

$$\begin{aligned}
G_{direct}^{VA} &= .50(10.779739) + .475(65.60) + .25(-3.60) \\
&\quad + 0(58.50) + .50(-42.00) + .55(-3.979739) \\
&= 12.461013.
\end{aligned}$$

The remaining direct federal/outside cost is \$72.838987 million. The nonmedical externality is \$1.485039 million; assigning 20 percent to Virginia adds \$0.297008 million of Virginia cost. Total Virginia/local net cost is then \$12.758021 million, leaving federal/outside net cost of \$36.927018 million.

D.3 Resident scenarios and the medical-saving crossing

Scenario	Virginia/local net cost	Federal/outside net cost	Resident cost	State-resident result	Status
Population $\lambda =$.0261299, no medical saving to Virginia	12.758	36.927	13.723	4.60	Derived, illustrative benchmark
Income-tax $\lambda =$.0288382, otherwise same	12.758	36.927	13.823	4.56	Derived, illustrative

Scenario	Virginia/local net cost	Federal/outside net cost	Resident cost	State-resident result	Status
Direct- budget case, all behavioral externali- ties excluded	12.461	72.839	14.364	4.39	Different component ledger
Treasury limit	12.758	excluded	12.758	4.95	Budget- boundary limit
Consolidated check	n/a	n/a	49.685	1.27	Calibrated to author result
25% of medical saving assigned to Virginia	3.483	46.202	4.690	13.45	Illustrative
50% of medical saving assigned to Virginia	-5.792	55.477	-4.342	pays for itself	Illustrative

The population benchmark divides Virginia’s population of 8,260,405 by the U.S. population of 316,128,839. The income-tax benchmark divides Virginia’s \$35.434 billion by the U.S. \$1,228.719 billion in 2013 federal individual income tax after credits.

The main jurisdictional uncertainty is the modeled \$37.1 million medical saving. Let s be the fraction assigned to Virginia, with the balance in the combined federal/outside account. Under the population weight,

$$D_{VA}(s) = 13.722919 - 37.1(1 - .0261299)s.$$

The denominator reaches zero at $s = 0.3798$, or about 38.0 percent. The income-tax weight gives a 38.4 percent crossing. The source assigns a dollar of long-run government medical saving to each dollar of childhood Medicaid/FAMIS spending by transporting estimates from another setting and extrapolating utilization through age 65. It leaves unidentified the child’s later residence, the public payer that saves, the future match rate, and whether FAMIS entrants have the same effect. The crossing is a jurisdictional sensitivity and carries no confidence-interval or forecast interpretation.

A second sensitivity substitutes the 18.7 percent general-fund share reported later for a broader Virginia eligibility-system portfolio for the benchmark 50 percent portal-development share. It also treats the TANF residual as fully controlled by Virginia. This gives Virginia/local net cost of about \$7.593 million, population-weighted resident cost of \$8.693 million, a ratio of about 7.26, and a medical-saving crossing near 24.1 percent. The evidence informs possible financing, though its broader portfolio is too distant from the 2012 CommonHelp portal to replace the benchmark.

Finally, the administrative-savings bars are modeled estimates without direct evidence of cash savings. Cholli and Wu estimate a proportional change in SNAP administrative cost per recipient and extend it to other programs. The \$42 million SNAP figure combines a graph reading with annual state-activity totals. These scenarios show which assumptions govern the result, while the available evidence supports no unique Virginia point estimate.

Appendix E. Additional participation cases

E.1 Massachusetts health-insurance automatic enrollment

Shepard and Wagner (2025) study Commonwealth Care applicants who had completed an application, been found eligible, and failed to choose a plan. Automatic assignment increased new enrollment by 48.4 percent relative to active enrollment alone. Suspending assignment reduced it by 32.6 percent relative to the automatic-enrollment baseline. The experiment therefore evaluates a post-eligibility default for people who had already applied.

The conventional welfare calculation uses one marginal passive-enrollee month as its unit. Insured medical claims are $C = 228.32$, modeled uncompensated care without insurance is $U = 135.313$, and enrollee WTP is $W = 93$. Shepard and Wagner assign 63.5 percent of uncompensated care to government and 36.5 percent to providers. Garthwaite, Gross, and Notowidigdo find that hospitals absorb approximately 60–67 percent, and 63.5 percent is their midpoint. Applying that evidence gives

$$G = .365U = 49.389, \quad P = .635U = 85.924.$$

The consolidated numerator is $W + P = 178.924$, while the denominator is $C - G = 178.931$, giving approximately one. Because $G + P = U$, reversing the split leaves this ratio almost unchanged while reallocating WTP and fiscal savings. That reallocation changes the state-resident calculation, where government savings and provider welfare enter differently.

Let m_C be the marginal federal share of coverage claims, and let s and f be the portions of the government uncompensated-care offset realized as state and federal cash savings. Let χ be the share of provider/private welfare accruing to Massachusetts residents, and let A_S, A_F, A_L be omitted implementation cost. Then

$$\begin{aligned} D_T &= (1 - m_C)C - sG + A_S + A_L, \\ D_R &= D_T + \lambda(m_C C - fG + A_F), \\ N_R &= W + \chi P. \end{aligned}$$

The enrolled months span gross federal shares of 50, 58.78, 60.19, and 61.59 percent. Early months fell under a capped waiver pool, while later eligibility could be treated under Title XIX. The paper reports neither the month-by-category weights nor whether a marginal federal claim displaced another waiver expenditure. The HSN uncompensated-care account mixed hospital assessments, private-payer surcharges, balances, waiver-related funding, possible state appropriations, and provider shortfalls. Thus, the coverage match cannot identify its marginal cash allocation.

For comparison with the main text, suppose the full government uncompensated-care offset is realized at the same federal/state shares as coverage. I set $A_S = A_F = A_L = 0$ and use Massachusetts's 2009 income-tax share, $\lambda = .032281$. This gives

Gross coverage match m_C	Treasury denominator	Resident denominator	Resident ratio, $\chi = 0$	$\chi = .5$	$\chi = 1$
0	178.931	178.931	0.520	0.760	1.000
.5000	89.465	92.353	1.007	1.472	1.937
.5878	73.755	77.150	1.205	1.762	2.319
.6019	71.232	74.709	1.245	1.820	2.395
.6159	68.727	72.285	1.287	1.881	2.475

The $m_C = 0$ row is a capped-pool cash-incidence stress test. If an added CommCare claim displaces another valuable service, complete welfare accounting must also value that service.

Using the population share, $\lambda = .021246$, gives 1.018/1.488/1.958 at a 50 percent match and 1.309/1.913/2.518 at a 61.59 percent match. These are derived conditional scenarios because the marginal match remains unidentified.

The HSN allocation also matters. Holding fixed the income-tax weight and source-consistent split gives

	Government HSN offset treated as	Resident denominator	Resident ratios for $\chi = 0, .5, 1$
50.00%	no cash offset	117.845	0.789 / 1.154 / 1.518
50.00%	same match as coverage	92.353	1.007 / 1.472 / 1.937
50.00%	all state/local saving	68.456	1.359 / 1.986 / 2.614
50.00%	all federal saving	116.251	0.800 / 1.170 / 1.539
61.59%	no cash offset	92.237	1.008 / 1.474 / 1.940
61.59%	same match as coverage	72.285	1.287 / 1.881 / 2.475
61.59%	all state/local saving	42.848	2.170 / 3.173 / 4.176
61.59%	all federal saving	90.643	1.026 / 1.500 / 1.974

The reviewed sources do not identify the marginal-offset row. The paper also excludes the state-only noncitizen group from its main sample, leaving no basis for a fully state-funded calculation.

At the printed WTP, the source-consistent consolidated result is already 0.99996, so any positive omitted implementation cost moves it farther below one. Adding 4 or 4.5 percent of claims as a scale proxy gives 0.951 and 0.946. These proxies come from aggregate Connector administrative fees; the marginal cost of automatic assignment is unmeasured. I therefore report a lattice over the coverage match, HSN allocation, and provider incidence. The evidence does not select a point from it.

E.2 Virginia’s ABAWD time limit

Gray et al. (2023) study Virginia’s October 2013 reinstatement of the historical three-month-in-36-month SNAP time limit for able-bodied adults without dependents. Near the age-50 exemption, the rule reduced 18-month retention among incumbent participants by 23.4 percentage points from a no-rule retention rate of 63.2 percent. Eliminating it changes legal eligibility, work or activity conditions, documentation, and recertification. The counterfactual therefore extends beyond application simplification.

The welfare calculation evaluates elimination of the rule in 2018Q1 dollars per relevant ABAWD-month. Its central ledger is

Component	WTP	Consolidated cost	Federal	Commonwealth	Virginia local
Restored SNAP benefits	74.47	74.47	74.47	0	0
Avoided new applications	0	-4.26	-2.13	-1.4697	-.6603
Added recertifications	0	1.84	.92	.6348	.2852
Net administration	0	-2.42	-1.21	-.8349	-.3751
Authors’ central earnings-related tax loss	0	11.03	11.03	0 benchmark	0
Additional bounded work-effort value	0 to 42.24	0	0	0	0

Component	WTP	Consolidated cost	Federal	Commonwealth	Virginia local
Central total	74.47 to 116.71	83.08	84.29	-.8349	-.3751

The restored-benefit term is $(.234/.632)(201.12) = 74.47$. Avoided applications save \$4.26, and additional recertifications cost \$1.84, for a net administrative saving of \$2.42. The \$11.03 tax term uses suggestive 24-month earnings results; the average at 18 months is statistically null. These inputs give a central consolidated range of 0.90–1.40. Setting the earnings-related tax effect to zero gives 1.03–1.40, while an adverse calibration gives 0.81–1.27.

SNAP benefits were federally financed. The historical administrative split was 50 percent federal, 34.5 percent Commonwealth, and 15.5 percent local. With no net Virginia income-tax response for the representative low-income filer, state and local government save \$1.21 per relevant month. The treasury account therefore **pays for itself on the measured components**.

Let λ be resident incidence of federal cost. With federal costs of \$73.26, \$84.29, and \$92.86 in the zero, central, and adverse earnings cases,

$$D_{VA} = -1.21 + \lambda G_F.$$

Earnings case	WTP	Resident denominator, population .0260812	Resident ratio	Denominator, tax share .0280391	Resident ratio
Zero	74.47–100.62	.7007	106.28–143.59	.8441	88.22–119.19
Central	74.47–116.71	.9884	75.35–118.08	1.1534	64.56–101.19
Adverse	74.47–116.71	1.2119	61.45–96.30	1.3937	53.43–83.74

The central denominator changes sign at $\lambda = .01436$. The reported 64.56–118.08 range combines two central denominators with the WTP endpoints. Since all of these ratios divide by roughly one dollar, the denominators provide the more stable information. The certification costs are back-of-the-envelope averages. Systems programming, notices, guidance, training, exemption screening, appeals, and work-program capacity are omitted from the calculation. A transition cost of one dollar per normalized unit would roughly double the central resident denominator.

The live Policy Impacts page reverses digits in \$74.47, describes retained incumbents as new participants, labels the 23.4-point retention effect as general participation, and reports 1.02 for the zero-earnings lower endpoint. The official online appendix gives 1.03, which is the value used here.

E.3 Mandatory SNAP Employment and Training for parents

Cook and East (2026) study parents already receiving SNAP in an unidentified mountain-plains state. A household head could be referred to mandatory Employment and Training when their

youngest child turned six shortly before recertification. The IV estimate implies that an induced referral reduces household SNAP benefits by \$669 over six months. The welfare counterfactual eliminates mandatory E&T and thereby restores \$111.50 per month. It assigns the recipient \$111.50 in WTP, sets the earnings-related fiscal effect to zero, and subtracts \$11.32 in planned average E&T cost. This gives

$$M_N = \frac{111.50}{111.50 - 11.32} = 1.113.$$

The \$11.32 consists of \$9.42 for supervised job search and \$1.90 for participant reimbursements. Historical rules divide the reimbursement line 50/50 between federal and nonfederal funds. Supervised job search could draw on the finite 100 percent federal formula grant, 50/50 reimbursement, or a mixture of the two. The paper leaves this mix unreported. A state, locality, workforce entity, community college, or nonprofit could provide the nonfederal match.

Let s denote the share of job search paid from the 100 percent federal grant. Let q denote the share of that grant expenditure that becomes an actual federal cash saving when E&T is eliminated. Federal and nonfederal savings are then

$$S_F = .95 + 4.71(1 - s) + 9.42qs, \quad S_N = .95 + 4.71(1 - s).$$

Next, let δ be the share of the nonfederal saving accruing to state residents. Using the historical zero direct state benefit share, the paper's WTP, and zero omitted fiscal and transition effects gives

$$D_R = \lambda(111.50 - S_F) - \delta S_N.$$

The core grid sets $q = 1$ and $\delta = 1$.

	All job search 50/50,		
λ	$s = 0$	Half formula, $s = .5$	All formula, $s = 1$
0	pays for itself, $D = -5.660$	pays for itself, $D = -3.305$	pays for itself, $D = -.950$
.005	pays for itself, $D = -5.131$	pays for itself, $D = -2.788$	pays for itself, $D = -.444$
.010	pays for itself, $D = -4.602$	pays for itself, $D = -2.270$	1,818.92, $D = .061$
.020	pays for itself, $D = -3.543$	pays for itself, $D = -1.235$	103.95 , $D = 1.073$
.050	pays for itself, $D = -.368$	59.65, $D = 1.869$	27.15, $D = 4.106$
.100	22.64, $D = 4.924$	15.83, $D = 7.044$	12.17, $D = 9.163$
1	1.11, $D = 100.180$	1.11, $D = 100.180$	1.11, $D = 100.180$

As s moves from zero to one, the zero crossings are $\lambda = .05348$, $.03194$, and $.00939$. At $\lambda = .02$, changing δ from zero to one moves the all-formula result from 55.13 to 103.95. If formula funding is fully reallocated, the cash-ledger denominator rises from 1.073 to 1.261 and the ratio falls from 103.95 to 88.42. A complete welfare calculation for this reallocation would also include the value of the grant’s alternative use. All of the very large ratios arise from denominators close to zero.

The \$11.32 comes from a later state plan and measures an average. The marginal saving per induced referral remains unmeasured, and the plan omits some staffing. The calculation also assigns zero to the value of avoiding the E&T ordeal, the private value of any lost voluntary service, labor-market fiscal effects, and transition costs. Adding \$5 of resident-weighted transition cost to the 103.95 scenario lowers it to 18.36; adding \$20 lowers it to 5.29. A point estimate would require the state plan’s funding columns and a consistent policy unit.

E.4 SNAP caseworker effectiveness

A separate paper by Cook and East (2025) uses quasi-random assignment of applicants in an anonymous mountain-plains state to SNAP caseworkers with different conditional approval rates. A one-standard-deviation increase in the caseworker index raises receipt in the application quarter by 2.68 percentage points and reduces incomplete applications by 1.27 points. These estimates are consistent with caseworkers helping complete applications that have already been started. The study leaves open which intervention, including training, staffing, software, or monitoring, would produce the one-standard-deviation shift.

For a hypothetical one-standard-deviation improvement, the unrounded three-year inputs per exposed applicant are

Component	Formula	Amount
Added benefits and paper WTP, B	$683(.03)$	20.4900
Added recipient-quarters	$2.657(.03)$	.07971
Routine administration, A	$(261/4)(2.657)(.03)$	5.2011
Earnings response, Y	$1,596(.03)$	47.8800
Paper fiscal saving	$.40Y$	19.1520
Net government cost	$B + A - .40Y$	6.5391
Reconstructed ratio	$20.49/6.5391$	3.13

The paper rounds the numerator to \$20, direct cost to \$25, and fiscal saving to \$19, then reports $20/(25 - 19) = 3.3$. Thus, 3.3 is the author-reported value and 3.13 is the unrounded reconstruction. The 40 percent fiscal rate combines a 24 percent SNAP phase-out, employee payroll tax, and a residual income-tax calculation with inconsistent filing conventions. The benefit outcome may already incorporate the SNAP phase-out, making the separate 24 percent offset a possible double count.

Historically, benefits were federal and ordinary administration was approximately 50/50. Under the paper’s taper convention, with zero implementation and state-tax effects, the accounts are

$$G_{SL} = 2.6005, \quad G_F = 3.9385.$$

Let K be implementation resource cost per induced completed application. Since the incompleteness effect is .01272 per exposed applicant, each \$100 of K adds \$1.272 before financing weights. Assuming 50/50 financing for implementation and routine administration gives the following grid of resident denominators and ratios.

	λ	$K = 0$	$K = 100$	$K = 500$
0, treasury limit		2.60 / 7.88	3.24 / 6.33	5.78 / 3.54
	.002	2.61 / 7.86	3.25 / 6.31	5.79 / 3.54
	.005	2.62 / 7.82	3.26 / 6.29	5.82 / 3.52
	.010	2.64 / 7.76	3.28 / 6.24	5.85 / 3.50
	.020	2.68 / 7.65	3.33 / 6.16	5.92 / 3.46
1, consolidated		6.54 / 3.13	7.81 / 2.62	12.90 / 1.59

The 6.29 reference uses $\lambda = .005$, full transfer valuation, a \$100 implementation cost per induced completion, no state tax effect, and a 50/50 implementation match. This is an illustrative calibration because the state and production technology remain unidentified. Halving transfer WTP and using $\lambda = .02$, $K = 500$ produces 1.73. With the same implementation cost entirely state funded, the ratio is 1.13. An implementable state-resident MVPF requires evidence on both the state and the intervention.

The live Policy Impacts page combines two papers. Its introduction describes the caseworker study, while the E&T worksheet, journal citation, and DOI come from the parents study. Every checked public version of the parents paper uses \$669 and \$111.50; the worksheet uses \$708 and \$118. The caseworker paper’s own welfare result is 3.3. The two policies must remain separate.

Appendix F. Source and version record

F.1 Inclusion and source rule

I include a study when it evaluates a U.S. public-transfer policy or participation mechanism, treats enrollment, retention, application completion, or continued participation as a primary outcome, and supplies either an MVPF or a sufficiently explicit welfare calculation. Cases with an incomplete policy boundary or state allocation remain in the additional-cases section, with the missing information stated directly.

The published article or dated working-paper version, its online appendix, and contemporaneous program-financing sources govern the calculations. I use Policy Impacts to discover cases and compare reported results, but resolve discrepancies using the underlying studies and authoritative program sources.

F.2 Included studies and versions

Study and policy or welfare counterfactual	Version used and local artifact	Policy Impacts page	Important version or transcription note
Finkelstein and Notowidigdo (2019), information and assistance for likely eligible Pennsylvania older adults	<i>Quarterly Journal of Economics</i> 134(3), 1505–1556; article PDF ; online appendix	Information; assistance	Information-page dollar fields are inconsistent; the assistance page omits the costless/resource-inclusive distinction
Unrath (2024), removing a quarterly CalFresh report	March 2024 manuscript; paper PDF	Removing SNAP Reporting Requirements	Figure 13 gives 4.25; printed page-23 inputs give 4.33; live library page retains older 2.09
Cholli and Wu (2026), Virginia CommonHelp	June 18, 2026 working paper distributed by the Washington Center for Equitable Growth in July 2026; paper PDF	No standalone page located as of August 18, 2026	Exact 1.27 and launch spending are reported, but aggregate WTP, direct-cost, and externality totals are graph-only and no numerical workbook was located
Shepard and Wagner (2025), Commonwealth Care automatic assignment	<i>American Economic Review</i> 115(3), 772–822; article PDF ; online appendix	Automatic Enrollment in Health Insurance	Paper and library invert the source direction of hospital versus government uncompensated-care incidence
Gray et al. (2023), eliminating Virginia’s ABAWD time limit	<i>American Economic Journal: Economic Policy</i> 15(1), 306–341; author manuscript ; official online appendix	Work Requirements for SNAP	Library has WTP, classification, and metadata errors; Online Appendix D.7 supplies the complete welfare ledger
Cook and East (2026), eliminating mandatory E&T for parents	May 27, 2025 author manuscript corresponding to <i>Journal of Public Economics</i> 257, article 105631; paper PDF	SNAP Caseworker Effectiveness, invalid combined record	Current public versions use \$669/\$111.50; the page uses \$708/\$118 and combines two papers

Study and policy or welfare counterfactual	Version used and local artifact	Policy Impacts page	Important version or transcription note
Cook and East (2025), hypothetical increase in SNAP caseworker effectiveness	NBER Working Paper 31307, revised May 2025; paper PDF	Same invalid combined record	Paper reports 3.3 and unrounded inputs give 3.13; the library’s E&T worksheet is a stale, unsupported splice from the parents calculation

F.3 Primary financing and incidence sources

I allocate government costs using the rules in effect during each evaluation. USDA Food and Nutrition Service program and administrative-cost sources provide the SNAP benefit and historical administrative shares. Contemporaneous Assembly Budget Committee and California Department of Social Services documents provide California’s 50/35/15 administrative division. HHS and CMS tables give the Virginia Medicaid/FAMIS match rates. Administration for Children and Families financial reports govern the TANF block-grant and maintenance-of-effort treatment. For Massachusetts coverage and HSN, I use waiver documents, Connector annual reports, and the HSN annual report.

Agrawal, Hoyt, and Ly (2024) provide the fiscal-federalism context for the local and social boundaries; the state-resident boundary is the separate incidence adjustment developed here. The exact [November 2024 paper PDF](#) used here is retained with the other source papers.

The paper-specific dossiers provide direct source URLs, page pinpoints, file hashes, and the complete arithmetic audits: Pennsylvania, California, CommonHelp, Massachusetts, Virginia ABAWD, Parents/E&T, and caseworkers.

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